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Enhancing Stewardship of Indigenous Peoples' Data, Information, and Knowledges in Libraries and Archives through Indigenous Data Governance

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Abstract

This article provides foundational definitions connected to Indigenous Peoples' data in relation to information institutions, specifically libraries and archives. The authors explain the relationship between Indigenous Data Sovereignty (IDSov) and Indigenous Data Governance (IDGov) before moving into a general overview of norms and principles related to enhancing IDSov and implementing IDGov and finishing with the CARE Principles for Indigenous Data Governance (Collective Benefit, Authority to Control, Responsibility, and Ethics). This article also provides examples of the application and

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implementation of the CARE Principles in libraries and archives, including Community-Driven Archives, Indigenous Librarianship, Local Contexts Labels and Notices in Libraries, and Language Materials. This article concludes with recommendations for applying the CARE Principles in information institutions. The purpose of the article's information is to provide foundational article, definitions, resources, and perspectives for information institutions and professionals, specifically librarians and archivists, to support IDSov and apply and use IDGov principles from Indigenous Peoples' viewpoints. The authors' intention is to show the importance of this work in communities, not just in information institutions. This article also highlights the importance of Indigenous librarianship when operationalizing IDGov. This article acknowledges the work being done in various information institutions to de-silo information practices and incorporate Indigenous Peoples' perspectives.

Keywords
Indigenous studies, Native Americans, American Indians, Indigenous Data Sovereignty, Indigenous Data Governance, CARE Principles

Indigenous Peoples and Libraries

This article provides an overview of issues involving Indigenous Peoples' data in information institutions before providing potential pathways to address them. To fully understand the challenges facing Indigenous Peoples related to their data, this article provides foundational definitions of Indigenous Peoples, Indigenous Peoples' data, information institutions, Indigenous Data Sovereignty (IDSov), Indigenous Data Governance (IDGov), and related principles and practices. By providing context, this article seeks to take into account how Indigenous information needs can be met through library and archival science. This article provides multiple examples within different information institutions showing implementation of IDGov. The information in this article is intended to provide an Indigenous Peoples' perspective and de-silo practices in information institutions.

Foundations of Indigenous Peoples' Self-Determination in Libraries

Who Are Indigenous Peoples?

Indigenous Peoples live on all continents throughout the world, representing as many cultures as places; as such, there is no single, unifying, or authoritative definition of Indigenous Peoples (United Nations Human Rights 2013). Instead, Indigenous Peoples are self-defined and include

such characteristics as historical continuity, distinctiveness, and a resolve to preserve, develop, and transmit their ancestral territories and identities according to their cultural patterns, social institutions, and legal systems (Cobo 1986; United Nations Human Rights 2013). The United Nations Permanent Forum on Indigenous Peoples has also stressed distinct language, distinguishable belief and culture systems, a strong link to natural resources and territories, and unique political, social, or economic systems (United Nations Human Rights 2013). Therefore, it is up to the individual community to self-identify as Indigenous and to define who belongs and their membership.

Throughout this article, it is also important to note that there are certain terms that are used interchangeably, depending on elements such as context or the quoted texts. These terms include "Indigenous Peoples," "Indigenous nations," and "Indigenous communities." "Native nations" and "tribes" are also used to refer to U.S.-based Indigenous Peoples. Additionally, the terms "settler-colonial" and "Western" are used to describe practices and information institutions in direct contrast with Indigenous values and principles.

What Are Information Institutions?

In this article, we use "information institutions" as an umbrella term for GLAM institutions (galleries, libraries, archives, museums), in addition to documentation centers, information centers, or any other entity collecting, classifying, preserving, providing access to, or stewarding Indigenous Peoples' information, data, records, and materials. Although contributors are coming from a U.S. perspective, the case studies and recommendations may apply to other CANZUS (Canada, Australia, New Zealand, and the United States) institutions.

Additionally, we differentiate between information institutions operating within settler-colonial settings versus Indigenous Peoples' own information institutions that include (1) Indigenous community and individually owned and operated institutions that may have non-Indigenous staff as well as (2) non-Indigenous institutions run by Indigenous leadership. Indigenous Peoples engage in a variety of discussions about the stewardship of their data across diverse information institutions including libraries and archives. It is important to note that from Indigenous Peoples' perspectives, their data in libraries are not siloed or independent from other information institutions. For Indigenous Peoples, there exists a similarity among strategies for implementing their rights, interests, and responsibilities across information institutions.

In the United States, Native nations operate tribal GLAM information institutions. In many cases, tribal libraries lack sustainable funding, staff, and resources (Roy and Alonzo 2003). This dearth of resources hinders

Indigenous Peoples from utilizing library and information sciences toward tribal nation rebuilding, meaning the information is difficult to use for strengthening a nation's capacity for culturally relevant and effective self-government as well as self-determined and sustainable community development (Native Nations Institute, n.d.; Duarte 2017). The lack of resources also impedes community awareness of the role of tribal librarians for cultural reclamation. Since tribal libraries can play a prominent role in fostering information literacy for community members, we note the potential for these institutions to educate tribal members on IDSov and IDGov through culturally appropriate information literacy and digital literacy (Roy, Bhasin, and Arriaga 2011).

Information institutions, particularly those in the GLAM fields, are often siloed by type due to the lack of knowledge on each other's role in cultural heritage work and general stereotypes of each profession. For example, pieces of literature within information sciences have typically dealt only with the handling of cultural materials within a single category of information institution such as archives or libraries. Within Indigenous communities, this is exponentially worse due to the lack of communication between tribal GLAMs despite the interdisciplinary nature of tribal librarians, archivists, curators, and cultural experts.

What Are Indigenous Peoples' Data?

Data are not foreign concepts to Indigenous Peoples because they have always been data creators, users, and stewards with data embedded in their instructional practices and their cultural principles (Garcia 2018; Carroll, Rodriguez-Lonebear, and Martinez 2019). For example, many Indigenous Peoples' knowledge systems were based on gathering data through generations via observation and experience that iteratively informed Indigenous Peoples' protocols, practices, and interactions with other people and the natural world (Rodriguez-Lonebear 2016; David-Chavez and Gavin 2018). In the United States, Indigenous Peoples' data have been recorded using oral histories, winter counts, stories, totem poles, calendar sticks, and other instruments storing and sharing information for the community's benefit (Rodriguez-Lonebear 2016). Indigenous Peoples' knowledges belong to them as collectives and include data informing Indigenous ways of knowing to ensure transmission to the next generations (Cajete 2000; Carroll, Rodriguez-Lonebear, and Martinez 2019). Indigenous Peoples' data also include Indigenous Peoples' data systems that rely on the knowledges, values, roles, and intergenerational responsibilities within communities to steward information and knowledge (Cajete 2000; Carroll, Rodriguez-Lonebear, and Martinez 2019). Finally, Indigenous Peoples' data include information and knowledges that protect and promote proper protocols for their use (Cajete 2000; Carroll, Rodriguez-Lonebear, and Martinez 2019).

Indigenous Peoples' knowledges and data were documented without consent by researchers in various disciplines, especially anthropology, into settler-colonial records and then held captive in information institutions (Deloria 1978; Anderson 2012; Krebs 2012). As part of this extractive documentation, examples of Indigenous data records were also taken from communities and can be viewed in a variety of settler-colonial information institutions. These projects of documenting and extracting Indigenous Peoples' data compose part of a larger strategy of settler-colonial erasure, co-optation, suppression, and control (Kukutai and Taylor 2016b; Rainie, Rodriguez-Lonebear, and Martinez 2017a). The pillage of Indigenous Peoples' Knowledges in the United States has led to five hundred years of non-Indigenous scholarly work, primary sources, and accumulated data, authorized and housed in non-Indigenous information institutions.

Indigenous Peoples' data can be born digital or emerge as tangible or intangible information, knowledges, languages, cultures, resources, materials, specimens, and objects (Rainie et al. 2017; Carroll, Rodriguez-Lonebear, and Martinez 2019). These data relate to Indigenous Peoples as collectives and/or individuals, including data about their nonhuman relations, citizens and community members, governments, and cultures (Nickerson 2017; Duarte et al. 2019; Lovett et al. 2019; Rainie et al. 2019; Carroll, Garba, et al. 2020). Indigenous Peoples' data include information collected by governments and institutions and are intrinsic to the communities' capability and capacity to realize their human rights, self-determination, and responsibilities to all of creation (Research Data Alliance 2019). Indigenous Peoples' conceptions of data highlight a more fluid boundary of data, information, wisdom, and knowledges to those defined in Western contexts that focus on digital characters, quantities, and symbols as data distinct from other forms (Carroll, Rodriguez-Lonebear, and Martinez 2019).

A broad range of data-related challenges exist for Indigenous Peoples. For instance, information institutions, repositories, and other online databases hold large collections of tangible and intangible cultural material, knowledge, and data (Carroll, Hudson, et al. 2020). The extensive information collected and held about Indigenous Peoples was rarely collected by or for Indigenous Peoples' purposes (Cannataci 2018; Walter 2018; Carroll, Rodriguez-Lonebear, and Martinez 2019). Consequently, much of the data do not recognize or privilege Indigenous Peoples' worldviews or benefit Indigenous Peoples (Cannataci 2018).

Another challenge is that Indigenous Peoples' data ecosystems have historically been characterized by several factors, including, first, external control and ownership of data or that Indigenous Peoples' data have not been owned or controlled by their communities (Kukutai and Taylor 2016a; Rodriguez-Lonebear 2016; Walter 2016; Rainie et al. 2017; Carroll, Rodriguez-Lonebear, and Martinez 2019). Second, data have

predominantly described Indigenous Peoples and their lifeways through a deficit lens. And third, there has been the presence of untrustworthy data actors—the people, places, and things that interact with data—resulting in irrelevant, inaccurate, or inconsistent data (Kukutai and Taylor 2016a; Rodriguez-Lonebear 2016; Walter 2016; Rainie et al. 2017; Carroll, Rodriguez-Lonebear, and Martinez 2019). For the purposes of this article, "data actors" refers to the information institutions and information professionals interacting with Indigenous Peoples' data. Fourth, there has been a lack of external support for Indigenous Peoples' data infrastructure and capability (Kukutai and Taylor 2016a; Rodriguez-Lonebear 2016; Walter 2016; Rainie et al. 2017; Carroll, Rodriguez-Lonebear, and Martinez 2019). Finally, there exists a mistrust by Indigenous Peoples of any data emerging from exploitative research and policies (Kukutai and Taylor 2016a; Rodriguez-Lonebear 2016; Walter 2016; Rainie et al. 2017; Carroll, Rodriguez-Lonebear, and Martinez 2019; Hudson et al. 2020).

Last, open access within settler-colonial institutions is coupled with settler-colonial narratives and practices that ignore the relationships and histories Indigenous Peoples have with these institutions, narratives, and practices. Although open access is valuable in many contexts, not all Indigenous Peoples' data are meant to be shared indiscriminately, so the implementation of IDSov frameworks would situate Indigenous Peoples' data back under their control and governance.

How Do Indigenous Data Sovereignty and Indigenous Data Governance Relate to Indigenous Peoples' Data, Information, and Knowledges?

The advancement of Indigenous Peoples' self-determination and the reclamation of Indigenous Peoples' knowledges and identity have led to the emergence of IDSov, an assertion of the rights, interests, and responsibilities of Indigenous Peoples in relation to data about them as individuals and collectives, their governments and cultures, and their nonhuman kin, including animals, the landscape, and other aspects of the natural world (Kukutai and Taylor 2016b; Snipp 2016; Rainie et al. 2017; Carroll, Rodriguez-Lonebear, and Martinez 2019; Carroll, Garba, et al. 2020). IDSov defines the right of Indigenous Peoples, nations, and communities to govern the collection, application, and use of data about their peoples, governments, and nonhuman relations, as well as reinforces the right of engaging in decision making in accordance with Indigenous Peoples' values and collective interests (Kukutai and Taylor 2016b; Research Data Alliance 2019). IDSov disrupts current data ecosystems, privileging Indigenous Peoples' paradigms to shift power dynamics and realize Indigenous Peoples' goals and visions for their data (Carroll, Rodriguez-Lonebear, and Martinez 2019).

Indigenous Peoples as sovereign collectives activate IDSov through

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the interrelated processes of decolonizing data and IDGov (Rodriguez-Lonebear 2016; Smith 2016; Walter et al. 2018; Carroll, Rodriguez-Lonebear, and Martinez 2019; Rainie et al. 2019; Carroll and Cummins 2024). Although not dealt with in this article, decolonizing data occurs when Indigenous Peoples replace non-Indigenous norms with Indigenous systems and frameworks that define data, how they are collected, who owns them, who has access, and how they are to be used into the future (Carroll, Rodriguez-Lonebear, and Martinez 2019). In tandem with decolonizing data, IDGov harnesses Indigenous Peoples' ways of knowing and doing and utilizes them (1) for the control and management of an Indigenous Nation's own data and (2) to inform how others steward Indigenous Peoples' data (Rainie et al. 2017; Walter et al. 2018; Carroll, Rodriguez-Lonebear, and Martinez 2019). IDGov combines community-held responsibilities with the authority to decide when and how Indigenous Peoples' data are collected, analyzed, accessed, stored, and/or used (Walter et al. 2018). IDGov can be described as a reciprocal relationship between data for governance and governance of data; the first deals with quality, relevance, access and use, and the second is about control and ownership (Carroll, Rodriguez-Lonebear, and Martinez 2019). At the center of this relationship is the exercise of sovereignty, including IDSov (Carroll, Rodriguez-Lonebear, and Martinez 2019).

For Indigenous Peoples, IDGov at its core involves being a good ancestor, collaborating with other data stewards, and establishing data-driven futures by Indigenous Peoples for Indigenous Peoples (Carroll, Rodriguez-Lonebear, and Martinez 2019). Relationships are also at the core of IDGov and with further development of Indigenous Peoples' own data governance mechanisms and principles, opportunities for strengthening nation-to-nation relationships between Indigenous and non-Indigenous data actors expand (Smith 2016; Carroll, Rodriguez-Lonebear, and Martinez 2019; Walter and Carroll 2021). Implementing IDGov within external data systems and with other data actors supports Indigenous Peoples' protection and control of data and increases access to, use of, and benefit from the data (Carroll, Garba, et al. 2022; Carroll, Plevel, et al. 2022).

To this end, Indigenous-led initiatives to enhance IDGov and create IDGov principles focus on the importance of acknowledging and promoting IDSov as well as leading with Indigenous core values, leadership, and scholarship in data governance (Kukutai and Taylor 2016a; Rodriguez-Lonebear 2016; Carroll, Rodriguez-Lonebear, and Martinez 2019; Carroll, Garba, et al. 2020; Carroll et al. 2021). International networks such as the Global Indigenous Data Alliance and the International Indigenous Data Sovereignty Interest Group at the Research Data Alliance bring together regional and nation-state networks including the First Nations Information Governance Centre, the Te Mana Raraunga–Māori Data Sovereignty

Network, the U.S. Indigenous Data Sovereignty Network, the Maiam nayri Wingara Indigenous Data Sovereignty Collective, the Pacific Data Sovereignty Network, and GIDA-Sápmi (see the appendix). These networks aid in promoting IDSov through the adoption of IDGov agendas and principles. Approaches to promote IDSov and implement IDGov must also include dialogue embracing multiple ways of knowing; support and utilize Indigenous Peoples' existing IDGov-related procedures and protocols; and conduct data, science, and research work in service to and with Indigenous communities (Carroll, Rodriguez-Lonebear, and Martinez 2019; Walter and Carroll 2021). Overall, any efforts enhancing IDGov toward IDSov require a commitment to listening to Indigenous Peoples in data decisions and any discussions affecting Indigenous Peoples, their communities, and nonhuman relations (Carroll, Rodriguez-Lonebear, and Martinez 2019).

One facet of IDGov is the use of and work by non-Indigenous allies who advocate for the construction of mutual, new, relational forms of dialogue, theory, research, and action (Kovach 2021). Non-Indigenous allies working within information institutions push back against settler-colonial practices and respect Indigenous methodologies and their applications (Kovach 2021). These advocates work to form relationships based on respect, equity, responsibility, and healing (Kovach 2021). Non-Indigenous allies can be key to endorsing and supporting the application and implementation of IDGov principles within information institutions. For example, non-Indigenous allies can identify the invisible labor that most often falls upon Indigenous shoulders and take proactive, coordinated, and well-communicated steps to move that labor to non-Indigenous allies. This can include establishing institutional IDGov working groups and cocreating IDGov documents and/or working strategies.

Norms and Principles for Indigenous Peoples' Data in Information Institutions

Through the weaving together and application of discourses involving international and national standards and Indigenous Peoples' expectations for research ethics and data practices, information institutions can honor Indigenous Peoples' individual and collective rights, identities, and knowledges by implementing IDGov principles. The following norms and principles are applicable to Indigenous Peoples' data in information institutions within the United States, although there are others applicable in other countries such as the First Nations Principles of OCAP (Ownership, Control, Access, Possession) in Canada (First Nations Information Governance Centre, n.d.) and the Principles of Māori Data Sovereignty (Te Mana Raraunga 2018). While some of the following norms and principles are specific to a type of institution, there is a disciplinary relationship between them and all information institutions and professions. GLAM

institutions can adapt and apply these norms and principles to meet their institutional needs for implementation of IDGov principles.

Universal Declaration of Human Rights (UDHR)

The UDHR as a human rights instrument informs various aspects of information practices, with two of the articles being key. First, Article 19 states, "Everyone has the right to freedom of opinion and expression; this right includes freedom to hold opinions without interference and to seek, receive and impart information and ideas through any media and regardless of frontiers" (United Nations General Assembly 1948). Second, Article 27 states, "(1) Everyone has the right freely to participate in the cultural life of the community, to enjoy the arts and to share in scientific advancement and its benefits. (2) Everyone has the right to the protection of the moral and material interests resulting from any scientific, literary or artistic production of which he is the author" (United Nations General Assembly 1948). Although not directly related to information institutions, UDHR does provide an avenue for information professionals to include human rights and contributes a foundation for thinking about and developing services and policies (Edwards 2010).

Despite its foundational principles, there are several limits in UDHR's application that violate or work against Indigenous Peoples' IDSov. For example, UDHR focuses on the individual rather than the community, assuming a single culture and a single community (Roy and Hogan 2010). This individualistic focus and generalization cause minority cultures such as Indigenous Peoples and their members to be continuously silenced. Additionally, with UDHR limited in focus on human rights, it is important for information professionals to also recognize Indigenous Peoples' cultures as the cultural objects themselves as well as the process through which the objects are created and used (Roy and Hogan 2010). Although UDHR may help provide an avenue for developing practices based on human rights, information institutions and professionals must build collections and practices reflecting various Indigenous Peoples' traditions, frameworks, and viewpoints that are attached to their data. Therefore, information professionals and institutions must be conscious of how strict adherence or guidance from this document causes information professionals and institutions to perpetuate settler-colonial practices and ideals to continue to silence Indigenous Peoples. The UDHR must be supplemented with the other norms and principles described next.

UN Permanent Forum on Indigenous Issues (UNPFII)

The UNPFII is an advisory body on Indigenous issues to the UN Economic and Social Council and uses an IDSov policy approach. Data and data collection have emerged as major concerns within the UNPFII, and various recommendations have subsequently been made to the UN and other

governing bodies (Davis 2016). The UNPFII has discussed work based on rights affirmed in the United Nations Declaration on the Rights of Indigenous Peoples (UNDRIP). Although the UNPFII has not completed work directly related to IDSov, its efforts generally attempt to bridge the information divide between Indigenous Peoples and other states within the UN system. The UNPFII also seeks to influence the international community through the consideration of development of culturally appropriate indicators reflecting Indigenous Peoples' worldviews (Yap and Yu 2016).

Despite UNPFII's work in relation to Indigenous Peoples, it does not yet specifically address Indigenous Peoples' data or its importance to Indigenous Peoples' sovereignty either generally or involving information institutions. Rather, the UNPFII focuses on a human rights framework to pursue its goals. While valuable as a potential avenue for implementation of other IDSov and IDGov principles in libraries and archives, the UNPFII's work would likely take on these issues as part of its human rights agenda rather than under the purview of sovereignty. This could include future side events at the UNPFII that focus on IDSov and supporting the translation and circulation of IDGov strategies developed by communities. This would directly assist in increasing information dissemination to Indigenous representatives to these meetings. A longer term goal for the UNPFII would be including Member State reports on the status of Indigenous data support and activity.

UN Declaration on the Rights of Indigenous Peoples (UNDRIP)

In confirming Indigenous Peoples' right to self-determination, UNDRIP necessarily affirms their right that data collected either by them or jointly with them reflect their past and present realities as well as provide the basis for their pursuit of self-determined social, economic, and cultural development (Tauli-Corpuz 2016). UNDRIP is a global consensus concerning Indigenous Peoples' rights, and although it is a nonbinding declaration, it sets forth a framework for state adoption in their relationship with Indigenous Peoples that may guide policy and protocol development (Davis 2016). As a human rights instrument, UNDRIP covers a range of rights. For example, Article 3 sets forth Indigenous Peoples' right to self-determination, which is tied to IDSov, and Article 11 mandates free, prior, and informed consent. Article 15 states that Indigenous Peoples have the right to the diversity of their cultures, histories, aspirations, and traditions, which must be reflected in public information and education (United Nations General Assembly 2007). Generally, UNDRIP emphasizes Indigenous Peoples' rights to strengthen and maintain their traditions, cultures, and institutions as well as pursue their own well-being in accordance with their own aspirations and needs (Kukutai and Taylor 2016a; Tsosie 2019).

Although UNDRIP provides a foundation for the advocacy and adoption of IDSov, because it is not a legally binding document, there are no consequences for not adopting its principles. UNDRIP is a state agreement, providing guidelines and recommendations for actions at the signatory level rather than an institutional one. Nevertheless, implementation of UNDRIP within signatory nation-states, local governments, and other institutions remains paramount to realizing responsibilities to IDGov generally as well as information institutions specifically.

Native American Graves Protection and Repatriation Act (NAGPRA)

In contrast to many of the other norms and principles mentioned, the Native American Graves Protection and Repatriation Act (NAGPRA; Pub. L. No. 101-601, 25 U.S.C. 3001-3013) is one of the few legally binding instruments at work in information institutions as it must be followed by any institution in the United States receiving federal funds. When passed, NAGPRA was a step forward in Indigenous Peoples' rights in its guidelines for the treatment, repatriation, and disposition of Native American human remains, funerary objects, sacred objects, and objects of cultural patrimony removed from tribal or federal lands. NAGPRA provides an untested possibility for the expression of Indigenous Peoples' rights in keeping with IDSov and IDGov principles. This is particularly in relation to data that may have been created in the course of institution and research study and analysis of NAGPRA-defined cultural items.

However, there are several shortcomings in NAGPRA regarding IDSov and IDGov principles. First, NAGPRA is legally binding only to institutions that have received federal funding. Second, NAGPRA does not include all aspects of Indigenous Peoples' data and is limited to four types of data as defined in the statute: human remains, unassociated funerary objects, cultural patrimony, and sacred objects. It is therefore up to individual institutions to adopt guidelines and protocols adhering to IDSov and IDGov principles, which include all aspects of Indigenous Peoples' data. Finally, although NAGPRA sets forth timelines, they can be lengthy, allowing institutions potential room for adhering to its provisions and thereby not following IDSov principles. For example, although an institution may have agreed to repatriate items back to Indigenous Peoples, scientists may delay the process out of a desire for continued research, especially in some emergent fields that include analysis of Ancient DNA. Another example is that photographs or other secondary data produced prior to NAGPRA through research on ancestors and held by research third parties are not necessarily included as part of repatriation practices. Again, the burden falls on Indigenous communities to know of the existence of this research and its current location in order to request full and comprehensive returns. Therefore, although NAGPRA is a step forward

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in Indigenous Peoples' rights and institutional adherence to IDSov and IDGov principles, it is not enough in and of itself and there are a range of loopholes that allow for third parties to keep data that should also be returned but are no longer in control of the institution subject to NAGPRA responsibilities.

Protocols for Native American Archival Materials

The Protocols for Native American Archival Materials (First Archivist Circle 2007) are built on the knowledge that Indigenous Peoples create, use, organize, and manage knowledge and information resources differently from settler-colonial institutions (Lawson 2004; Underhill 2006). The Protocols are a guide for information institutions engaging in culturally responsive care of Indigenous Peoples' materials and providing culturally appropriate service to all communities. With the understanding that each nation, community, and people is unique, the Protocols and best practices should be interpreted and applied by each institution and community (Underhill 2006; First Archivist Circle 2007). Although focused on archives, the Protocols set forth standards relevant to any and every information institution.

The Protocols support IDSov and IDGov by focusing on Indigenous Peoples' perspectives when stewarding their information. The Protocols also focus on relationship building with Indigenous Peoples whose materials are housed within an institution. This relationship is fundamental to IDSov and IDGov principles. Relationship building brings Indigenous Peoples' perspectives into any conversation involving the stewardship of their data. However, the Protocols are often misinterpreted as pertaining only to Indigenous Peoples' data held in archival institutions but should be implemented in any information institution holding Indigenous Peoples' data as a way to implement or adopt IDSov and IDGov principles. Although the Protocols show how to enact IDGov principles for Indigenous Peoples' data, they have not been widely adopted since being formalized in 2007. The Protocols also do not address all IDGov principles such as the importance of Indigenous Peoples' materials for their governance practices. The Protocols also apply more to cultural materials already housed in institutions rather than the processes of how those materials and data are gathered, collected, and used in the first place.

American Philosophical Society Protocols

The practices of the American Philosophical Society (APS) Center for Native American and Indigenous Research (CNAIR) in Philadelphia provide a strong example of institutional compliance with IDSov. The leadership of APS in this area of library practices is also significant because the history and size of its collections make APS the oldest and one of the largest

repositories of materials on Indigenous languages, histories, and culture in North America (Carpenter 2021).

The APS Protocols for the Treatment of Indigenous Materials (APS Protocols)—adapted in 2014—formalize ways and means of how to balance Indigenous Peoples' sovereign rights with (1) the legal obligations of the APS library to donors and (2) the APS mission to provide free access to its materials for scholarship and education. The APS Protocols formalize the APS objective to co-steward Indigenous Peoples' materials and data with representatives from the community where the data originated. The APS Protocols guide the processes of co-stewardship by providing recommendations in three interrelated areas: (1) assessment of materials for sensitive content and determining proper ways of managing these materials, (2) relationship building with Indigenous Peoples and donors on proper treatment of Indigenous content, and (3) assistance APS provides to Indigenous Peoples in publishing materials (Powell 2014).

The APS Protocols resulted from the three years of collaboration among the APS librarians' team and the APS's Native American Advisory Board. The preceding partnerships between APS and Indigenous communities that began in 2008 during the Digital Knowledge Sharing initiative that led to the establishment of the center (Powell 2014) also contributed to the construction of this document. Since adoption of the APS Protocols in 2014, the APS CNAIR has worked with over eighty Indigenous communities throughout North America on sharing digital copies of materials, collaboratively describing and representing content, and protecting culturally sensitive content (Carpenter 2019). Its functioning guides other institutions on how to handle sensitive Indigenous content (Newberry Library 2021). The APS Protocols also may be used as a model of constructing policies in collaboration with communities to both formalize a library's commitment to IDGov and as a way to support Indigenous collection management in ethically and culturally responsive ways.

As valuable as the APS Protocols may be to information institutions, it should be noted that they are often interpreted as being from a settler-colonial perspective rather than an Indigenous one. While the APS Protocols may provide some guidance on pursuing IDSov within information institutions, they should be implemented in conjunction with other norms and principles to ensure Indigenous viewpoints and frameworks are used for IDGov application of Indigenous Peoples' data.

CARE Principles for Indigenous Data Governance

The CARE Principles for IDGov (Collective Benefit, Authority to Control, Responsibility, and Ethics; see figure 1) were distilled from discussions involving frameworks informing processes for stewardship and control of Indigenous data in areas such as research methods, research governance,

and science that reflect Indigenous Peoples' rights and cultures (Cajete 2000; Anderson, Griew, and McAullay 2003; Castellano 2004; Wilson 2008; McGregor, Restoule, and Johnston 2018; Carroll, Garba, et al. 2020; Carroll, Hudson, et al. 2020; Kovach 2021). The CARE Principles are a people- and purpose-focused framework that empowers Indigenous Peoples by focusing on values-based relationships promoting equitable Indigenous participation in data reuse processes to result in more equitable outcomes involving Indigenous data as well as preserving relationships of trust and respect (Carroll, Hudson, et al. 2020). Ultimately, the high-level CARE Principles refer data actors back to the communities from which the data originate to define standards and protocols to steward, access, use, and benefit from data (Carroll, Garba, et al. 2020).

The CARE Principles express the minimum expectations for all data actors, including all institutions and repositories collecting, analyzing, storing, using, and reusing Indigenous Peoples' data (Carroll, Garba, et al. 2020; Carroll and Cummins 2024). As data actors advance practices and standards facilitating data use, reuse, and research reproducibility, the CARE Principles have the goal of creating value from Indigenous Peoples' data for them while realizing opportunities for Indigenous Peoples within data ecosystems (Carroll, Hudson, et al. 2020). Complementing the FAIR Principles (Findable, Accessible, Interoperable, Reusable) (GO FAIR, n.d.), the CARE Principles encourage data movements, producers, users, and repositories to consider people and purpose in advocacy and pursuits (Research Data Alliance 2019; Carroll, Hudson, et al. 2020). The CARE Principles are therefore designed to be implemented in tandem with the FAIR Principles by data stewards and users (Carroll, Garba, et al. 2020). In implementing the CARE and FAIR Principles structurally and operationally, it will be important to create consistency by developing standards, criteria, and guidelines (Carroll, Hudson, et al. 2020).

Although each principle is conceptually distinct, the CARE Principles relate to each other and define concepts, rights, and interests to be employed and used when facilitating Indigenous Peoples' control in their data governance and reuse (Carroll, Garba, et al. 2020). Although information institutions can implement a single principle or multiple simultaneously, IDGov must address all CARE Principles to effectively uphold Indigenous Peoples' rights (Carroll, Garba, et al. 2020).

Examples: CARE in Information Institutions

Community approaches provide a path toward implementation of the CARE Principles in information institutions, especially libraries and archives. The CARE Principles emphasize management of Indigenous Peoples' data in response to Indigenous Peoples' protocols guiding management of Indigenous Peoples' data located outside of their communities. The following examples provide instances of potential avenues of

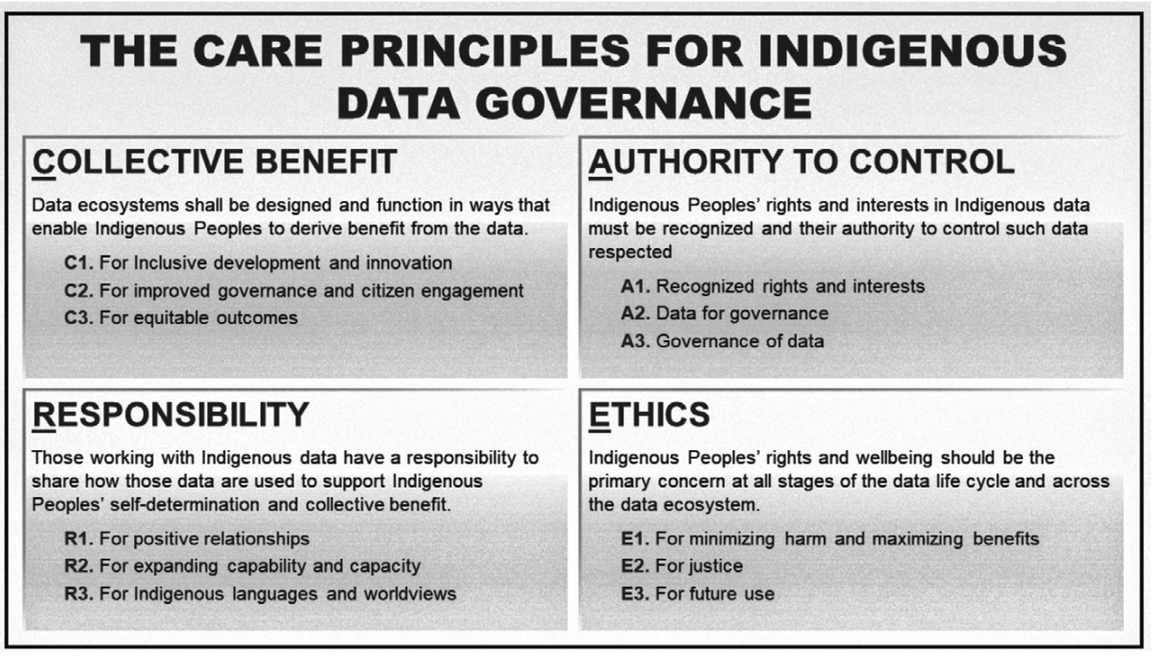


Figure 1. CARE Principles and subprinciples.

implementing the CARE Principles to Indigenous Peoples' data in information institutions, particularly libraries and archives. It should be noted that every institution's implementation of the CARE Principles depends on contextual factors, constraints, and overall commitment to transfer the control of Indigenous information and knowledges to Indigenous Peoples.

Example 1: Role of Community-Driven Archives in Indigenous Data Sovereignty and Governance

For Indigenous Peoples, contemporary memory keeping of photographs, audiovisual material, and documents can seem foreign or unimportant for cultural reclamation due to the harm and knowledge extraction done by settler-colonial information institutions in collecting those items. Indigenous Peoples are often unaware of how archives can protect and pass down cultural heritage. Observations of Indigenous farming techniques chartered down in journal entries, homemade Indigenous language teaching tools written by grandparents for cultural transmission, and audiovisual recordings documenting family history all represent Indigenous knowledge and information. But the historic silos perpetuated by settler-colonial information institutions and academia have isolated such information according to terms like data, stories, archives, collections and research. With the emergence of Community-Driven Archives (CDA) methodology, Native nations in Arizona have begun to utilize the CDA framework for the development of community archival repositories that move beyond this Western definition of knowledge preservation since it empowers Indigenous community members to define it, not the profession nor the academy. This is mainly because CDA methodology seeks

to reimagine and transform twenty-first-century libraries and archives by dismantling racist power structures that lead to erasure and trauma for Black, Indigenous, and other People of Color and LGBTQ communities in the record (Godoy 2021).

In practice, CDA methodology leads to the empowerment of community archivists to tell and steward their own information/knowledge. By educating community members on core archival practices such as appraisal, arrangement and description, and preservation, CDA praxis empowers community members to reimagine what an archive can be within their own community (Godoy 2021). This power shift to enhance community archives has paved a path for Indigenous Peoples to insert Indigenous values and perspectives within contemporary memory-keeping approaches and practices. Its emphasis on community definitions of appraisal, arrangement and description, and preservation has led Indigenous practitioners of CDA to begin to expand applications to also include community definitions of stewardship and ownership. Additionally, knowledge of library and archival resources that restrict access based on community protocols, like the Mukurtu content management system (CMS), gives community members critical knowledge on how to utilize the library and information science field as a means for cultural sovereignty, meaning the ability to protect and defend Indigenous Peoples' cultures to ensure cultural survival (Coffey and Tsosie 2001).

By becoming aware of the possibilities, Indigenous community archivists are able to articulate and apply their community protocols to govern their archives and begin to advocate for the critical information infrastructure needed to sustain it. Although nested in the world of archives by non-Indigenous information institutional perspectives, CDA methodology aligns with IDSov due to its emphasis on community ownership and stewardship of its memory. By educating community members on core archival practices, then encouraging them to Indigenize it to meet their needs, meaning to include, adapt, or incorporate Indigenous ways of knowing and frameworks into archival practice, Indigenous CDA seamlessly enacts IDGov.

Example 2: Labriola National American Indian Data Center

As information institutions aspire to adopt IDGov principles within their policies and procedures, a clear indicator of their commitment to safeguard and defend Indigenous Peoples' information and knowledge can be seen in their recruitment and retention of Indigenous information professionals such as librarians and archivists (Cooper et al. 2019). Within Western information institutions, Indigenous librarians and archivists can assert localized application of Indigenous librarianship to support Indigenous data stewardship and incorporate Indigenous cultural protocols within collections that have historically been stewarded only by non-Indigenous

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professionals (Cooper et al. 2019). As defined by Kathleen Burns et al., Indigenous librarianship is "grounded in the contemporary realities of Indigenous Peoples and Indigenous aspirations for self-governance and sovereignty, it has a critical theoretical base" (2018, 2331). Its practice may be carried out in places ranging from small Indigenous community libraries to specialized collections in large research institutions. In contrast to the neutrality of settler-colonial librarianship, which argues that information professionals are neutral actors in the collection, usage, and portrayal of information, Indigenous librarianship seeks to provide culturally relevant library and information collections and services by, for, and with Indigenous Peoples. With the emergence of Indigenous librarianship throughout the world, the recruitment and retention of Indigenous information professionals are critical to combat information historically collected, curated, and disseminated by non-Indigenous information professionals. The extraction of Indigenous knowledge by Western information institutions has created a power differential that subjugates Indigenous stories and data to neocolonial databases and repositories that do not honor Indigenous protocols pertaining to Indigenous knowledge. Indigenous librarianship stands in contrast to Western librarianship practices and the Open Access movement by enforcing IDGov when mediating information by restricting information access to uphold Indigenous cultural protocols. Non-Indigenous information professionals do not have the cultural fluency and cultural humility to advocate for the relatives and ancestors' residing within data, nor do they understand the context of the information being mediated to know when or what type of restrictions to place on the information. Indigenous protocols should govern all information pertaining to Indigenous knowledges (histories, cultures, science, and heritage) held within Western information institutions. In library or archival settings, the integrity of Indigenous Peoples' data, information, and knowledges is protected only when Indigenous librarians and archivists are serving as information mediators or intermediaries.

When meaningfully and sustainably supported by information institutions, Indigenous librarianship provides a practical mechanism to enact IDSov and operationalize IDGov principles to support the cultural sovereignty of tribal communities. One emerging example of Indigenous librarianship in action is Arizona State University's (ASU) Labriola National American Indian Data Center (Labriola Center). After the 2021 appointment of the Labriola Center's first Indigenous director in its twenty-eight-year history, the Labriola Center has grown fourfold with an all Indigenous staff that includes two Indigenous librarians and two Indigenous program coordinators. This Indigenous library within the ASU Library system seeks to empower Indigenous Peoples in the pursuit of resistance, persistence, and prosperity through the support of Indigenous research, scholarship, cultural expression, memory keeping, and community

learning. Under the mantra "Information is Sacred," Labriola staff value Indigenous definitions of kinship, community, consent, and sovereignty throughout the Labriola Center service model, specifically within its collection stewardship and research services (Soto 2023). With institutional support from senior ASU Library leadership, senior ASU Indigenous leadership working directly with the university's president, and ASU Indigenous faculty in various disciplines, the Labriola Center has deployed a community-engaged and collaborative approach to the stewardship of cultural knowledge that enhances IDSov and is synonymous with IDGov principles (Labriola, n.d.). With the increase in Indigenous staff, specifically with the hiring of a new Labriola Librarian, the Labriola Center is implementing the Protocols for Native American Archival Materials (First Archivist Circle 2007) throughout the ASU Library system (Sanchez et al. 2013). Currently, the Labriola Center is surveying the library's collection for compliance with the Protocols and working with ASU's Office of American Indian Initiatives to establish consultations with Native Nations regarding ASU Library material that may need to be repatriated to tribes. Concurrently, the Labriola Center is educating non-Indigenous ASU Library colleagues on how to handle research inquiries that involve Indigenous Peoples' knowledges and information, with the goal of instituting a rigorous ethical standard protecting the integrity of this data throughout ASU's entire library department. Additionally, the Labriola Center is employing the Protocols beyond archival material under the conviction that circulating books (fiction and nonfiction), scholarly journals, dissertations and theses, audiovisual material, maps, and data sets should be stewarded to the same standard as archival material.

With the increase of staffing capacity at the Labriola Center, non-Indigenous ASU librarians and archivists can respectfully collaborate with Labriola staff in a timely manner to explore research policies and procedures supporting IDGov and to enact the CARE Principles. For example, the Labriola Center is working with the Open Science & Scholarly Communication (OSSC) team within the library to develop library policies and procedures tailored to the unique information needs of Indigenous Peoples and Indigenous researchers. Due to the degree of cultural fluency and cultural humility needed to customize and localize IDGov, the Labriola Center and OSSC team have worked collaboratively on data management plans submitted by scholars and faculty to best embed language supporting Indigenous ownership and stewardship of data. Through this process, the Labriola Center and OSSC are identifying ways that IDGov and CARE Principles can be implemented within the research data cycle. With the presence of Indigenous librarians within an academic library, the Labriola Center seeks to normalize Indigenous care practices for all library data inquiries.

Outside of the university, the Labriola Center's community-engaged

and collaborative approach to foster cultural sovereignty among local and regional tribes in Arizona, specifically in O'odham communities, is exemplified through Labriola's Indigenization of CDA theory and praxis (Godoy 2021). As members of tribal communities as well as information professionals, Labriola Center leadership views CDA as a methodology rather than an information service; it is a new way of thinking Labriola aims to impart. Since 2019, the Labriola Center has worked on reimaging the archives through culturally informed community workshops that demonstrate how archives document contemporary Indigeneity and teach that community and family archivists are modern memory keepers. Through the distribution of professional-quality photo preservation supplies, technical assistance with family-produced oral history recordings, scanning photographs into preservation standard file formats, and preservation of various ephemeral materials, the Labriola Center has at an individual scale linked long-held community concepts such as community ownership, stewardship, reciprocity, and mediation to information science bedrocks such as the previously discussed norms and principles. When a CDA workshop is facilitated by Indigenous Peoples, this creates a setting for discussions about the importance of Indigenous control of information and knowledge, the validity of Indigenous knowledge, and the need for intergenerational learning environments (youth to elders). In all workshop outreach, Labriola Center staff promote community control of the history and data held within community collections and do not ask for any material to be donated to the university. Rather, the Labriola Center engagement is focused on educating and empowering Indigenous community archivists to steward and develop collections within the community, utilizing and reinforcing the inherent power and authority Indigenous Peoples have. The role of Labriola Center staff is to provide archival consultations, resources, and community-specific training so tribal members can reimagine how to preserve, organize, and mediate their information through CDA methodology.

When the work of consultations, resource referral, and training is scaled up, CDA methodology can be used for self-determination and tribal sovereignty. This is seen in the Labriola Center's partnership with the Hia-Ced O'odham Tribe. Currently, the Hia-Ced O'odham Tribe are not federally recognized because they were falsely deemed extinct by anthropologists despite having a prominent presence in southern Arizona and being recognized by nearby tribes and federal governmental agencies (Bell, Anderson, and Stewart 1980; Martinez 2013, 2022; Hudson et al. 2023). After attending Labriola Center CDA workshops, the Hia-Ced Hemajkam O'odham LLC (standing governmental body of the Hia-Ced O'odham) are now utilizing CDA methodology to develop their community archives into evidentiary proof of existence for the federal acknowledgment process. Instead of donating their archival material to the information

institution of ASU, they are using CDA resources of the Labriola Center to organize, preserve, and mediate their material, enacting sovereignty on their own terms (Pereira 2021). As the saying goes, iron sharpens iron, and in working with the Hia-Ced O'odham, at times in remote areas of the desert with scanners powered by gas generators, the collaboration with the Hia-Ced O'odham has added a place-based facet to CDA methodology of being where the memory is, whenever possible. Labriola Center leadership is training community members to process their own material, and throughout consultation the Labriola Center has honored their right to govern their information and has redirected university resources to support the Hia-Ced O'odham pursuit of federal recognition (Whitby 2022).

Once informed and empowered to understand the scope of archival infrastructure needs, Hia-Ced O'odham expressed a need to create a community archival repository that best reflects their O'odham ways of knowing. Specifically, they noted the need to develop O'odham metadata description for their collection and the means to provide culturally appropriate access points for their community's history and culture. As an Indigenous library on O'odham land, the Labriola Center is mindful of the territory it resides on and seeks to center it when applicable. Due to the degree of capacity needed to pursue their request, the Labriola Center partnered with the Hia-Ced O'odham for a million-dollar grant from the Andrew W. Mellon Foundation. Awarded in 2023, this three-year grant seeks to build archival frameworks and legal agreements to protect Indigenous intellectual property and to codevelop archives that adhere to Indigenous cultural sovereignty. Titled "Firekeepers: Building Archival Data Sovereignty through Indigenous Memory Keeping," this initiative will enact a community-based participatory archives approach in order to foster archival data sovereignty, meaning the ability for tribal communities to control the telling, gathering, and preservation of their knowledge positioned in the archives. This will consist of a community consultation process that will assist in the bridging of Indigenous ways of knowing and Western academic concepts of knowledge organization and affirm intergenerational local community understandings of information governance and archival data stewardship. One of the key deliverables is to institute the Mukurtu CMS for Arizona tribal use as well as provide formal legal archival template agreements between libraries and tribal rightsholders that honor and center Indigenous intellectual property rights and cultural protocols and implement the use of Traditional Knowledge Labels within Mukurtu.

Although large in scope, the Labriola Center now has foundational capacity to pursue IDSov and IDGov initiatives internally as well as outside the university. This would not be possible without the commitment from ASU Library to prioritize Indigenous librarianship and to staff the

expansion of the Labriola Center. By prioritizing the recruitment and retention of Indigenous librarians, non-Indigenous information institutions can begin to rebuild and establish trust with Indigenous Peoples in hopes of correcting settler-colonial narratives of extinction. Through this process, Indigenous librarianship and specifically Indigenous CDA provide practical points of entry for tribal members to actualize IDSov within their communities.

Example 3: Local Contexts Labels and Notices in Libraries

On May 4, 2018, the Library of Congress released the first online catalog record displaying the TK Labels, a digital marker and an innovative decolonial intervention enabling the recognition of Indigenous political authority over cultural heritage held in museums, archives, and libraries around the world. Celebrating this historic transition for digital research infrastructures and for Indigenous communities, Dwayne Tomah from the Passamaquoddy Tribe in Maine stood at a podium at the Library of Congress with tears streaming down his cheeks, singing a song that had been lost to the community for 128 years. The song was once again publicly heard and its rightful connection to the Passamaquoddy Tribe recognized and affirmed in the catalog record, finding aids, and metadata. In a Local Contexts film reflecting on this moment, Dwayne says, "We are still here. . . . It's amazing" (Local Contexts 2021). Embedding these TK Labels in the official record meant that, for the first time, proper attribution to the Passamaquoddy community was made, thereby changing forever how these recordings will be remembered publicly, used by the community, and employed by researchers well into the future (Maloney 2018).

The Passamaquoddy experience is not an isolated one. The erasure from collections and cultural heritage institutions is shared by Indigenous communities across the world. Every Native American, Alaskan Native, and Hawaiian Native community has cultural and biological collections within national archives, libraries, and museums that they do not own, do not control, and cannot govern care and circulation over. Significant information about these vital collections, including individual and community names and proper provenance information, is often known or can be ascertained but is strikingly absent. Issues of responsibility, inventorship, ownership as well as the incomplete and/or significant mistakes in the metadata extend to every other knowledge asset or digital record building upon this information. Recognizing Indigenous rights in cultural heritage, intellectual property, and genetic resources is difficult to achieve through our current legal mechanisms because of limits imposed by intellectual property law (e.g., preexisting knowledge, tangible expression, length of time). This tension is exacerbated by increasing digitization since there is no legislation or policy in place that protects Indigenous Peoples' digital knowledge sharing. Ultimately, this disregard for Indigenous rights affects

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cultural memory, the accuracy of historical narratives, and present-day Indigenous culture, health, and well-being and is also a critical matter pertaining to Indigenous knowledge and data sovereignty.

How to build equity, diversity, and sovereignty into digital infrastructures is a critical issue of our time. Indigenous cultural heritage and biological collections are unique in composition, in content, in their social and cultural value to the communities from where they derive, and to non-Indigenous publics seeking to better understand the complexity of Indigenous cultures and cultural practices. While these collections are enormous, they unfortunately have limited information, leading to impoverished metadata due to settler-colonial practices of data collection (Anderson and Christen 2019; Carroll, Hudson, et al. 2020). For example, between 1875 and 1957 one collector, Gustav Heye, assembled a collection of 225,000 catalogue records representing over 700,000 objects and over 500,000 photographs (Greenwald and Smith 2015). This collection now forms the basis of the National Museum of the American Indian. In an example of the governmental focus on documenting Indigenous lives and practices, the American Indian Records Repository (AIRR) is physically an underground facility holding over 200,000 indexed boxes, each box holding one cubic foot of material or approximately 2,500 sheets of paper (Greenwald and Smith 2015). While these boxes hold historical records from as far back as the 1700s, the AIRR continues to receive boxes from retired Bureau of Indian Affairs offices from around the country, with a monthly average of boxes received being 1,157 (Greenwald and Smith 2015). Cultural heritage was removed from communities and detached from local knowledge systems. For example, the National Anthropological Archive holds over eighteen thousand cubic feet of cultural heritage material including one million ethnological photographs, 11,400 sound recordings, and six thousand hours of film (National Museum of Natural History, n.d.). The American Folklife Center at the Library of Congress also holds over six thousand hours of sound collected on Indigenous lands (G. Shankar, pers. comm., May 15, 2021). In addition, complex historical-political conditions create ethical concerns around access and future circulation. As noted above, organizations like the Society of American Archivists have established specific protocols for the treatment of Indigenous collections generally and increasingly libraries and archives around the globe have heeded Indigenous Peoples' calls to integrate Indigenous curatorial models and knowledge into mainstream library and archive practices, from cataloging to display modes.

Local Contexts was founded in 2010 to bring forth the voices, authority, and local power of Indigenous communities in sharing and managing their intellectual and cultural property in cultural heritage, biological, and genetic resources. Now a 501(c)(3) nonprofit registered in the Navajo Nation, Local Contexts delivers the TK and BC Labels and Notices as an

extralegal provenance, protocol, and permission mechanism establishing equity by embedding Indigenous rights and interests within digital infrastructures. It offers digital strategies for Indigenous communities, cultural institutions, and researchers and has two primary objectives (Prehn et al. 2023):

- To enhance and legitimize locally based decision-making and Indigenous governance frameworks for determining ownership, access, and culturally appropriate conditions for sharing historical, contemporary, and future collections of cultural, biological, and genetic heritage and associated data
- To increase Indigenous involvement in data governance through the integration of Indigenous values into data systems and metadata structures

Local Contexts delivers the unique innovation of the TK and Biocultural (BC) Labels and Notices and Hub provision and delivery service (Local Contexts 2021). Developed over ten years in collaboration with Indigenous community partners around the world, Labels and Notices address historical legacies of exclusion within records and data, building new relationships of trust and collaboration around data use and potential commercialization opportunities for the future (Local Contexts 2021). TK and BC Labels and Notices are digital tags that correct records, bring Indigenous rights and interests into institutional contexts, and educate non-Indigenous Peoples about Indigenous rules and responsibilities around sharing knowledge (Montenegro 2019). TK and BC Labels and Notices have been embedded within key national digital infrastructures including at the U.S. Library of Congress and are a foundation of the new Indigenous Data Provenance Standard being developed with the international standard setting organization IEEE (Local Contexts 2021). The impact of digital labeling using the TK and BC Labels and Notices is foundational as well as far-reaching, helping to address (Local Contexts 2021)

- erasure of Indigenous culture and contributions in educational settings like museums, archives, libraries, and universities;
- unethical extraction of Indigenous knowledge in national parks and on Indigenous lands and waters;
- promotion of IDSov and IDGov;
- building of a global digital infrastructure for research, innovation, and historical inquiry that is far more equitable and inclusive;
- national obligations under the United Nations Declaration on the Rights of Indigenous Peoples, which includes sharing of the benefits that derive from Indigenous knowledge and data.

The complementary digital tagging system of TK and BC Notices has been developed and adopted by researchers and institutions to identify

and make more visible and transparent Indigenous collections and Indigenous contributions often hidden within information institutions as well as other cultural, research, and scientific organizations. For example, the APS in Philadelphia has integrated these Notices at multiple levels of their digital infrastructure (see CNAIR 2024). The Notices are also used in the APS's *Guide to the Indigenous Material at the American Philosophical Society* (2024) as a way of engaging in new decolonial practices that include publicly acknowledging and committing to new processes for incorporating and recognizing Indigenous rights and interests in cultural collections.

The TK and BC Labels are delivered through the Local Contexts Hub. The Hub, based at New York University (NYU) Libraries, is an open-source project that enables information exchange through an open API. The Hub is a safe place for communities to customize their unique Labels and then deliver them directly to archives, libraries, museums, and data repositories. The Hub allows for distinct community control over the customizing of the text for each label. In this way each community retains control and authority over their Labels, can update them, and can deliver them to institutions as needed. The Hub is also a space for institutions and researchers to apply TK and BC Notices and to support the inclusion of Indigenous rights and interests into their own institutional contexts. The Hub utilizes the power of open persistent identifiers for Labels and Notices to enable machine learning and enhanced search capabilities.

Example 4: Language Materials

More attention has been devoted to managing sensitive content than to languages themselves, but uncontrolled access to language items can be problematic for a variety of reasons. Languages, like physical objects, exist within relationships and practices of knowledge circulation that often involve controls on the circulation of words, sounds, ways of speaking, or languages as a whole (Debenport 2010). Relevant considerations include orthographic choices, purism against borrowings from other languages, and cultural restrictions of language that can be heard, read, or used appropriately only by certain individuals or in certain contexts. Many Indigenous communities engaging in language revitalization have worked to shape those efforts to limit the circulation of their languages both within and outside their communities (Hill 2002; Debenport 2010).

As such, implementing the CARE Principles in language archives requires attention not only to sensitive content but also to language itself. This challenge is reflected in the online dissemination of John Peabody Harrington's research papers. Harrington, one of the twentieth century's most prolific linguists, transcribed more than a million pages of famously inscrutable notes of words in western North American Indigenous languages (University of California, Davis 2022). Despite their difficulty, Harrington's notes have proven extremely useful for speakers of Chochenyo,

Kitanemuk, Juaneño, and other languages (Moore 2006). Though incomplete, the project to digitize and annotate Harrington's notes, based at the University of California, Davis (UC Davis), has made available several microfilm reels of notes. With accessibility a priority, access controls were evidently discussed only regarding "gossip or personal information" and "material deemed by communities to be sacred, or culturally inappropriate for dissemination," with no consideration for proscriptions on the circulation of language itself (Macri, Golla, and Woodward 2009).

In contrast to the UC Davis–based project, the Mukurtu CMS incorporates cultural protocols for access not only to digital heritage items or data but also for language entries in its Dictionary service (Mukurtu 2020a). Mukurtu is not a standalone archive but rather a CMS for use by communities to govern the sharing of their cultural heritage. Mukurtu began as a platform for implementing protocols governing access to digital heritage items and data encompassing entries for various forms of tangible and intangible cultural heritage but has recently developed a Dictionary feature enabling the creation of communities of language stewards and contributors able to add entries for words and link them with related digital heritage items and data (Mukurtu 2020d). Metadata fields for dictionary entries currently include media assets, alternate forms and spellings, sample sentences, and links to related content (Mukurtu 2020a). Dictionary entries can be linked to digital heritage items and data, whose metadata includes fields for high-level category descriptors, individual or group creators, historical or social context, and TK Labels (Mukurtu 2020b). Through these multidirectional links to digital heritage items and data, dictionary items can be associated with communities, which are Mukurtu's primary means of implementing cultural protocols for access based on seasonal, gender, ceremonial, or other controls (Mukurtu 2020c). As such, Mukurtu enables fine-grained specification of the circulation of language entries that is absent in other online archival resources like the Harrington Database project. Such flexibility is just as important for language archives than for those devoted to other forms of cultural heritage.

The Mukurtu CMS adheres to the CARE Principles by centering Indigenous Peoples' perspectives through the integration of their authority to control their data with various access restrictions. Through this control and integration of Indigenous Peoples' perspectives, Mukurtu shifts Indigenous Peoples' data away from settler-colonial influences to better reflect their worldviews and frameworks.

Applying the CARE Principles in Information Institutions

The following applications and recommendations for implementation of the CARE Principles to information institutions, specifically libraries and archives, can be adapted and adopted based on individual situational and

institutional factors. The following items often center bringing an Indigenous information professional into settler-colonial information institution practices. The following items also work to address the de-siloing of institutional practices and should be adopted within information institutions to honor Indigenous Peoples' individual and collective rights, data, identities, and knowledges. As noted above, arbitrary decisions made by non-Indigenous information professionals negatively impact how Indigenous Peoples' knowledges are handled in the Western information ecosystem. If handled by Indigenous information professionals, the integrity of the data or story can be best ensured through Indigenous librarianship since it takes into account IDSov. However, although there are no legal or extralegal consequences for not implementing IDSov or IDGov principles in information institutions, non-Indigenous information institutions may practice IDGov by honoring Indigenous Peoples through practices such as application of the CARE Principles and ensuring the inclusion of an Indigenous Peoples' perspective in information institutions.

Indigenous Librarianship

The worldwide movement to advance Indigenous librarianship centers Indigenous Peoples' perspectives in all information inquiries and institutional policies and practices pertaining to Indigenous knowledges. All information institutions, not just libraries, can practice or advance Indigenous librarianship; however, Western information institution professionals, specifically non-Indigenous librarians and archivists, often have not yet realized the responsibility to support Indigenous librarianship. Advancing IDGov through Indigenous librarianship requires Western information institutions to recruit, hire, retain, and support Indigenous librarians and archivists, preferably ones who originate from the lands the Western information institution resides on, in critical library positions that handle Indigenous stories and data. Within Indigenous communities, Indigenous librarians and archivists themselves become agents of change and actors in applying IDSov for the care, protection, and use of Indigenous knowledges. Within any information institution, Indigenous librarianship provides a mechanism for enacting and operationalizing IDSov and IDGov by weaving Indigenous community-specific protocols and the CARE Principles with Indigenous knowledges in library and archival operations.

Community-Driven Archives

Within the library and archival professions, the emerging methodology of CDA offers a reparative approach for Indigenous Peoples' memory keeping, which includes stories and data found in family and community archival material (Godoy 2021). Additionally, it centers community ownership and stewardship, not ownership and stewardship by Western information

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institutions, in all aspects of memory keeping. In Arizona, Indigenous community archivists have highlighted how CDA can decolonize and Indigenize archival practices for cultural reclamation and cultural sovereignty. They also have highlighted how Indigenous librarians and archivists working within a Western information institution have found ways to begin to align institutional values to support Indigenous archival praxis, IDSov, and IDGov. CDA enacts the CARE Principles through the establishment of culturally sensitive archival practices and protocols that support tribal self-determination and Indigenous ownership of knowledge and information. This inclusiveness to center Indigenous protocols highlights how CDA methodology can shift the power dynamics within Western information institutions when handling or stewarding Indigenous knowledges.

Enrich Metadata

Metadata attached to Indigenous Peoples' data at information institutions too often do not reflect the sovereignty of Indigenous Peoples but rather reflect the settler-colonial practices of the Western information institution itself. The usage of such tools as Local Contexts Labels and Notices, Mukurtu CMS, cultural protocols, and Indigenous languages enriches the metadata of Indigenous Peoples' data to digitally embed provenance, attribution, permissions, protocols, and more. Whether an information institution uses tools such as TK Labels, the metadata of Indigenous Peoples' data are enriched and their sovereignty is honored. Using these metadata tools, the CARE Principles are being incorporated into the settler-colonial practices of Western information institutions as Indigenous Peoples' perspectives, frameworks, rights, and interests are embedded within digital infrastructures.

Culture Keeping

Historically, settler-colonial information institutions have been conducting institutional stewardship for the sake of collecting rather than community stewardship for cultural reclamation. This distinction means the difference between extractive practices for information institutions versus honoring Indigenous Peoples' sovereignty and data. A shift to community stewardship for cultural reclamation means that Indigenous Peoples will have a greater understanding of how information institutions, specifically libraries and archives, can protect and pass down cultural heritage contained in data. With a greater understanding of the role Indigenous information professionals play in culturally appropriate mediation of Indigenous Peoples' data, the creation and development of Indigenous-led information institutions is imperative for culture keeping as well as IDSov. Indigenous information professionals will be best at advocating for community use of library and archival material for cultural reclamation and

will have the cultural fluency to best communicate with tribal government and cultural practitioners the role of Indigenous librarianship. When Indigenous Peoples lead information institutions, their potential for community stewardship of culturally appropriate information literacy and digital literacy increases, fostering a culturally safe space within academia to enact IDSov and IDGov. Inherent with Indigenous-led libraries and archives or Western institutions focusing on the incorporation of permanent Indigenous staff, this inclusion of the Indigenous perspective to the framework and policies of the institution honors the CARE Principles by embedding Indigenous Peoples' perspectives, frameworks, and worldviews.

Non-Indigenous Allyship

Non-Indigenous allyship can be practiced or promoted by any non-Indigenous and Indigenous information professional or institution dedicated to promoting IDGov and the CARE Principles. By advocating for and promoting the development and use of non-Indigenous allies, information institutions can broaden available resources advocating for IDGov and the CARE Principles within all facets of information institutions. Cultivating these allyships also advances Indigenous frameworks and methodologies through the formation of relationships based on responsibility, respect, equity, and healing (Kovach 2021). Non-Indigenous allies can identify and take steps to implement IDGov and combat the amount of labor that often falls on Indigenous information professionals.

Conclusion

Although not expected to be an all-encompassing guide, this article conveys the importance of adopting IDSov, IDGov, and the CARE Principles in information institutions, specifically libraries and archives. The definitions, resources, explanations, and examples are meant to fill in the gaps in this new field of study to provide a greater understanding of IDSov. The norms and principles described in the article are to be understood and adopted as steps toward the recognition and adoption of protocols and practices that uphold IDSov; none of the information provided should be understood as an end goal for information institutions. The examples are evidence of potential initial steps to further IDSov and IDGov.

The adoption and application of IDSov, IDGov, and the CARE Principles in information institutions is an ongoing process. Institutions and information professionals need research and publication on the adoption of IDGov practices in information institutions, specifically libraries and archives, in order to share examples of possible adoption strategies. Most importantly, further work should elaborate on how the inclusion of Indigenous perspectives in settler institutions, through relationship building, employment practices, or institutional support, aids in the integration and normalization of IDGov practices.

APPENDIX. RESOURCES AND LINKS.

Resource	Link
Te Mana Raraunga Māori Data Sovereignty Network	https://www.temanararaunga.maori.nz
Global Indigenous Data Alliance	https://www.gida-global.org
International Indigenous Data Sovereignty Interest Group at the Research Data Alliance	https://www.rd-alliance.org/groups/intern ational-indigenous-data-sovereignty-ig
U.S. Indigenous Data Sovereignty Network	https://usindigenousdatanetwork.org
The Maiam nayri Wingara Indigenous Data Sovereignty Collective	https://www.maiamnayriwingara.org
Pacific Data Sovereignty	https://pacificdatasovereignty.com
First Nations Information Governance Centre	https://fnigc.ca
GIDA-Sápmi	https:/uit.no/research/sshfno/project?p_document_id=674134&pid=788403

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